Protocol design elements and USDM

Data or words?

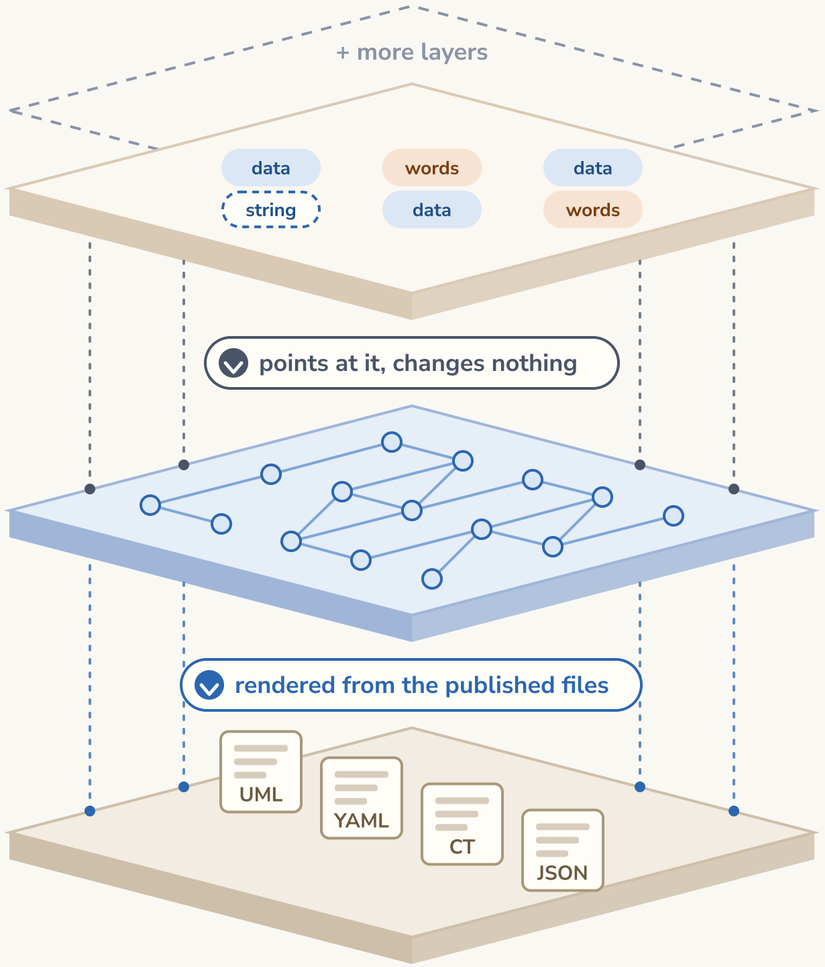
What USDM holds for objectives, endpoints, eligibility criteria and the schedule

Data | Data in a string | Words

Kerstin Forsberg · the published USDM v4 model and its CDISC example files, represented as a knowledge graph

Three layers

From the published model to readings on top



ON TOP

usdm-overlay ↗

Readings of what each attribute and class holds: data, data in a string or words, each with its evidence. Findings about the model, each with a test.

The overlay is one example. With the model as a knowledge graph, new layers can be added on top: other standards linked to it and queried together.

IN THE MIDDLE

usdm-rdf ↗

The same model, represented as a knowledge graph: a global identifier for each of the 86 classes and 693 attributes, linked to NCIt. Study files in JSON can be read into the same graph.

AT THE BASE

USDM v4, as published by CDISC ↗

The UML model, the same model as YAML, the terminology workbook (USDM_CT.xlsx) and the API specification. Study designs are exchanged as JSON files; the few public example files are not fully consistent or populated.

The question

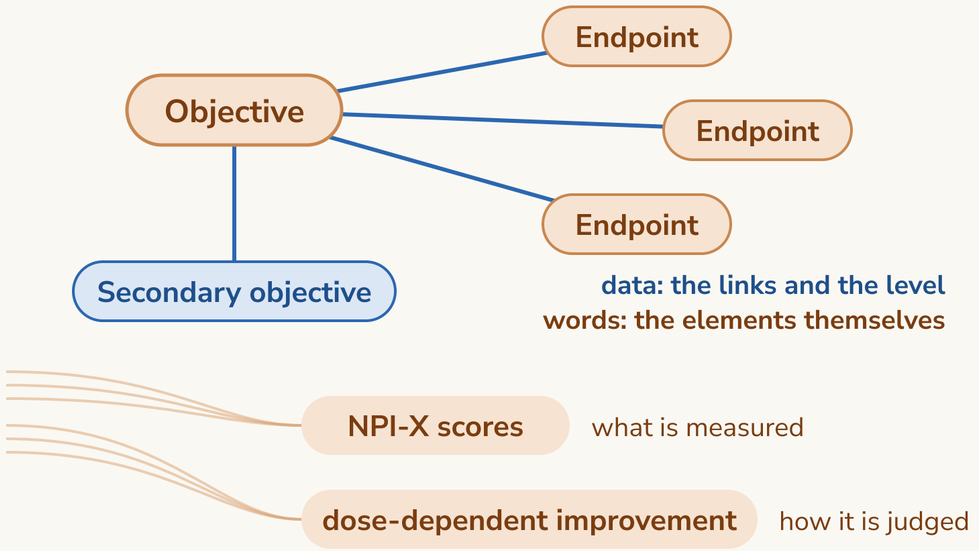
For each design element: which parts are data, and which are still words?

AN OBJECTIVE IN USDM · CDISC PILOT STUDY EXAMPLE

Level: secondary objective | Its 3 endpoints, linked

> “To assess the dose-dependent improvement in behavior. Improved scores on the Revised Neuropsychiatric Inventory (NPI-X) will indicate improvement in these areas.”

What is measured, and how it is judged, sits inside the sentence.



Data

Choices from a list, numbers, links. A system can check and reuse them.

Data in a string

An identifier or a duration written as text. Usable only if you know the convention.

Words

Sentences and written rules. A person has to read them.

These readings are the overlay, the top layer on slide 2. Objective as given in the CDISC Pilot example file (DDF-RA v4.0.0); the few public example files are not fully consistent or populated.

Five design elements and the protocol: the schedule is data, the rest is mostly words

Data | Data in a string | Words

Objective ↗

- Level
- Its endpoints
- The objective, written

Endpoint ↗

- Level
- The endpoint, written
- Its purpose

Eligibility criterion ↗

- Inclusion or exclusion
- Order
- Label like “I1”
- The criterion, written

no link to their sections

Protocol document version, sections ↗

no link in USDM

Study design

Schedule of activities ↗

	Visit	Visit	Visit	Visit
Activity	●	●		●
Activity		●	●	●
Activity	●			●

- visit type, contact mode
- what is measured

on slide 3: inside a sentence

- timing and windows
- Conditions for branching
- Visit start and end rules

Arms × epochs ↗

	Screening	Treatment	Follow-up
Arm 1	element	element	element
Arm 2	element	element	element

Inside each element: what happens, start rule, end rule

In short

USDM gives every design element an identity. Much of the content is still words.

Identity, for the objective on slide 3

Published USDM, the kind	name `Objective` · C-code `C142450` (Study Objective)
Published USDM, this one	id `Objective_3` · name `OBJ3` · unique within the study file
usdm-rdf adds	a global identifier for the kind, linked to NCIt C142450: `https://w3id.org/cdisc/usdm/v4/Objective`

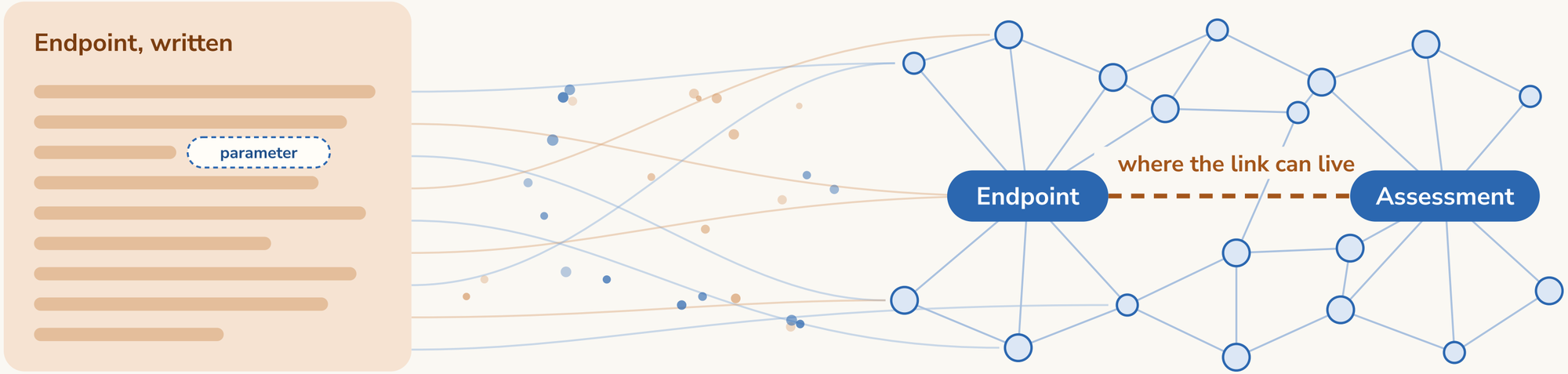
Held as data	The schedule of activities, and the arm and epoch grid
Data in a string	Timing values and windows; labels like “I1”
Held as words	Objectives, endpoints, eligibility criteria, rules inside elements
Not held	Endpoint to assessments; design element to protocol section

Beyond USDM

The words are not a dead end

A knowledge graph can represent the entities and relationships from study design to data capture. Structuring what the words say in such a graph turns them into data, and that is where the link from an endpoint to its assessments can live.

USDM already allows some structure inside the text: objective, endpoint and criterion texts are defined as structured text, and syntax templates can mark parameters and refer to model instances with markup. The few public example files use this little and are not fully consistent or populated; in the CDISC Pilot, no objective or endpoint text and 4 of 31 criterion texts carry such markup.



Such a link is one of the further layers that slide 2 leaves room for.